

Final Review

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Phase 3 · Final Review

Overview

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11 Phase 3 final review questions

Question 1

In the given equation, k is a constant. The equation has exactly one real solution. What is the value of k ?

$$x^2 + (\sqrt{k-3})x + 42 = 0$$

- A. 171
- B. 168
- C. 165
- D. 45

Answer 1

Correct Answer: A

Question 2

In the given system of equations, m and b are negative constants. In the xy -plane, the graphs of the equations in the given system intersect at the point $(-5, y)$, where $y < 0$. Which expression represents the value of b ?

$$x^2 + y^2 = 36$$

$$y = mx + \frac{b}{4}$$

A. $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$

B. $\frac{5m}{4} - \frac{\sqrt{11}}{4}$

C. $-20m + 4\sqrt{11}$

D. $20m - 4\sqrt{11}$

Answer 2

Correct Answer: D

Question 3

A biologist mixed a solution that is 0.3% sodium chloride by mass with a solution that is 0.15% sodium chloride by mass to obtain a new solution, which has a mass of 80 grams and contains 0.21 grams of sodium chloride. How many grams of 0.3% sodium chloride solution did the biologist use?

- A. 0.14
- B. 20
- C. 60
- D. 79.86

Answer 3

Correct Answer: C

Question 4

The function v is defined by the equation

$$v(x) = \frac{x^2 + bx + c}{(x + 5)(x - 19)},$$

where b and c are constants. In the xy -plane, the graph of $y = v(x)$ passes through the points $(0, \frac{66}{95})$ and $(11, 0)$. If $v(q) = 0$, which of the following could be the value of q ?

- A. -6
- B. -5
- C. 5
- D. 6

Answer 4

Correct Answer: A

Question 5

$$\frac{x}{5} + \frac{y}{9} = \frac{47}{45}$$

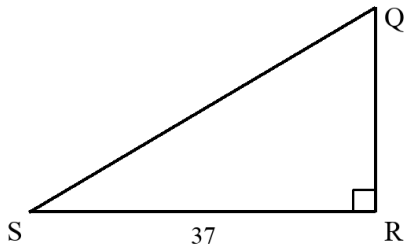
An engineer connects resistors in series, where the resistors in the series have a total resistance of $\frac{47}{45}$ ohms. In this series, there are x resistors of type A, which each have a resistance of a ohms, and y resistors of type B, which each have a resistance of b ohms. The given equation represents this situation. According to this equation, what is the positive difference between the value of a and the value of b ?

- A. 47
- B. 4
- C. $\frac{47}{45}$
- D. $\frac{4}{45}$

Answer 5

Correct Answer: D

Question 6



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

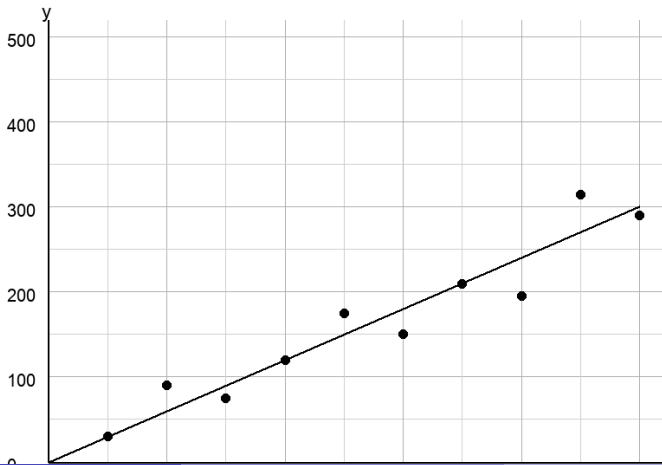
- A. $37 \cos Q$
- B. $37 \sin Q$

Answer 6

Correct Answer: D

Question 7

A group of 10 gardeners recorded data on the germination rates of their tomato crop for one growing season. The scatterplot shows the relationship between the number of tomato seeds planted, x , and the number of tomato seeds that germinated, y , for each of the gardeners. A line of best fit is also shown.



Answer 7

Correct Answer: C

Question 8

A rectangular banner has an area of 3,200 square inches. A copy of the banner is made in which the length and width of the original banner are each increased by 20%. What is the area of the copy, in square inches?

- A. 3,220
- B. 3,240
- C. 3,840
- D. 4,608

Answer 8

Correct Answer: D

Question 9

The mass of object A is 444% of the mass of object B, and the mass of object A is 0.740% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of p ?

Answer 9

Final Answer:

Question 10

In 2013, Chloe earned 11% more than in 2012, and in 2014 Chloe earned 6% more than in 2013. If Chloe earned y times as much in 2012 as in 2014, which of the following is closest to the value of y ?

- A. 0.5455
- B. 0.6600
- C. 0.8499
- D. 1.1766

Answer 10

Correct Answer: C

Question 11

Point F lies on a unit circle in the xy -plane and has coordinates $(1, 0)$. Point G is the center of the circle and has coordinates $(0, 0)$. Point H also lies on the circle and has coordinates $(-1, y)$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

- A. $\frac{37\pi}{2}$
- B. $\frac{39\pi}{2}$
- C. 34π
- D. 35π

Answer 11

Correct Answer: D

Thank You!

We appreciate your time and focus.

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